

Linear Models Lecture 5: Finite Sample Properties

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Why use $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ in my work?

- Motive 1: best approximation to the conditional expectation function (CEF).
 - While the conditional mean $E(Y|\mathbf{X})$ is the best predictor of Y , its form is typically unknown.
 - The linear model is an approximation to the conditional mean that has the lowest mean squared error among linear predictors.
- Motive 2: Sampling Properties.
 - When the underlying CEF is linear and the residual variance is constant, OLS is optimal.
 - OLS is the Best Linear Unbiased Estimator, here meaning the minimum variance (most efficient) linear estimator for the parameter vector β .

Why Study Finite Sample Properties?

- Finite sample results hold *exactly* for any sample size n —they do not require $n \rightarrow \infty$.
- The finite sample framework answers four questions:
 - 1 Is $\hat{\beta}$ centered on β ? $\rightarrow$ **Unbiasedness**.
 - 2 How much does $\hat{\beta}$ vary across samples? $\rightarrow$ **Variance**.
 - 3 Can any other estimator do better? $\rightarrow$ **Efficiency (Gauss-Markov)**.
 - 4 What if $\text{var}[\mathbf{e}|\mathbf{X}] \neq \sigma^2 \mathbf{I}$? $\rightarrow$ **GLS**.
- These results provide the theoretical foundation for standard errors, confidence intervals, and hypothesis tests.
- They also reveal *when* OLS is and is not optimal, motivating the alternatives we study in this lecture and beyond.

Finite Sample Assumptions of OLS estimator

- Assumption 1: The random variables $\{(Y_1, X_1), (Y_2, X_2), \dots, (Y_n, X_n)\}$ are independent and identically distributed.
- Assumption 2: $Y = \mathbf{X}'\beta + e$, where $\mathbb{E}(e|\mathbf{X}) = 0$.
- Assumption 3: $\mathbb{E}(\mathbf{X}\mathbf{X}') > 0$ is invertible (with probability 1).
- Assumption 4*: $\mathbb{E}[e^2|\mathbf{X}] = \sigma^2(\mathbf{X}) = \sigma^2$.

When Are These Assumptions Reasonable?

- **A1 (i.i.d.):** Holds well for cross-sectional surveys with random sampling. Fails with time-series data (serial correlation), clustered data (students within schools), or panel data.
- **A2 ($\mathbb{E}[e|\mathbf{X}] = 0$):** Requires that no omitted variable is correlated with $\mathbf{X}$. Fails when there is selection bias, simultaneity, or measurement error in $\mathbf{X}$. This is the assumption most contested in applied work.
- **A3 (rank condition):** Fails with perfect multicollinearity (e.g. including a dummy for every category plus an intercept). Nearly violated when regressors are highly collinear.
- **A4* (homoskedasticity):** Rarely holds exactly. Variance of earnings typically grows with education; variance of GDP growth differs across countries. Violation does *not* bias $\hat{\beta}$, but makes OLS inefficient and standard errors wrong.

Goals for Sampling Properties.

- **unbiased**: The (conditional) expectation of the estimator $\hat{\beta}$ of β is equal to the parameter. $\mathbb{E}[\hat{\beta}] = \beta$
- **efficient**: The estimator $\hat{\beta}$ of β has a lower variance matrix than other estimators.
- The latter efficiency result is called the Gauss Markov Theorem and depends on Assumption 4*.

Unbiasedness

- We will show that $\mathbb{E}[\hat{\beta}|\mathbf{X}] = \beta$ using three methods: summation, matrix, and decomposition.
- Each relies heavily on the conditioning theorem:

$$\mathbb{E}[g(\mathbf{X})Y|\mathbf{X}] = g(\mathbf{X})\mathbb{E}[Y|\mathbf{X}]$$

and

$$\mathbb{E}[g(\mathbf{X})Y] = \mathbb{E}[g(\mathbf{X})\mathbb{E}[Y|\mathbf{X}]]$$

Unbiasedness of the OLS slope estimator: Version 1

- Assume $\mathbf{x}_i$ is a $k \times 1$ vector representing observation i
- $\mathbb{E}[Y_i | \mathbf{x}_1, \dots, \mathbf{x}_n] = \mathbb{E}[Y_i | \mathbf{x}_i]$ because of independence of observations across i .

Unbiasedness (shown in summation operation)

$$\begin{aligned}\mathbb{E}[\hat{\beta} | \mathbf{x}_1, \dots, \mathbf{x}_n] &= \mathbb{E} \left[\left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \right)^{-1} \left(\sum_{i=1}^n \mathbf{x}_i Y_i \right) \middle| \mathbf{x}_1, \dots, \mathbf{x}_n \right] \\ &= \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \right)^{-1} \mathbb{E} \left[\left(\sum_{i=1}^n \mathbf{x}_i Y_i \right) | \mathbf{x}_1, \dots, \mathbf{x}_n \right] && \text{(Cond. Theorem)} \\ &= \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \right)^{-1} \sum_{i=1}^n \mathbb{E}[\mathbf{x}_i Y_i | \mathbf{x}_1, \dots, \mathbf{x}_n] && \text{(Linearity)} \\ &= \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \right)^{-1} \sum_{i=1}^n \mathbf{x}_i \mathbb{E}[Y_i | \mathbf{x}_i] && \text{(Cond. Theorem + indep.)} \\ &= \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \right)^{-1} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \beta && \text{(Linear CEF)} \\ &= \beta && \text{(Inverse)}\end{aligned}$$

Unbiasedness of the OLS slope estimator: Version 2

Using Matrix notation, $\mathbb{E}[\mathbf{y}|\mathbf{X}] = \mathbf{X}\beta$, $\mathbb{E}[\mathbf{e}|\mathbf{X}] = 0$ (by assumption 2)

$$\begin{aligned}\mathbb{E}[\hat{\beta}|\mathbf{X}] &= \mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}|\mathbf{X}] \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbb{E}[\mathbf{y}|\mathbf{X}] && \text{(Conditioning Theorem)} \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta && \text{(Independence)} \\ &= \beta && \text{(Inverse)}\end{aligned}$$

Unbiasedness of the OLS slope estimator: Version 3

Decomposition, noting $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$

$$\begin{aligned}\hat{\boldsymbol{\beta}} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \mathbf{e}) && \text{(plug in)} \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e} && \text{(distribute)} \\ &= \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e} && \text{(Inverse)}\end{aligned}$$

So now we can just check that $\mathbb{E}[\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}|\mathbf{X}] = 0$ by applying assumption 2.

$$\mathbb{E}[\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}|\mathbf{X}] = \mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}|\mathbf{X}] = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbb{E}[\mathbf{e}|\mathbf{X}] = 0$$

Variance of Least Squares Estimator

We can write the variance of the OLS estimator in terms of

$$\mathbf{D} = \begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \sigma_n^2 \end{pmatrix}$$

Where σ_i^2 gives us the variation in the regression error for observation i .

Derivation of Variance of $\hat{\beta}$

$$\begin{aligned}\text{var}[\hat{\beta}|\mathbf{X}] &= E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'|\mathbf{X}] \\&= E[((\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}) - \beta)((\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}) - \beta)'|\mathbf{X}] \\&= E[((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e})((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e})'|\mathbf{X}] \\&= E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}\mathbf{e}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}|\mathbf{X}] \\&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\mathbf{e}\mathbf{e}'|\mathbf{X}]\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{D}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\end{aligned}$$

Note that $\mathbf{X}'\mathbf{D}\mathbf{X} = \sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i\sigma_i^2$

The Sandwich Formula in Practice

- The variance $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{D}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$ is called the **sandwich formula**: $(\mathbf{X}'\mathbf{X})^{-1}$ is the “bread” and $\mathbf{X}'\mathbf{D}\mathbf{X}$ is the “meat.”
- This is the formula behind every “robust standard error” you see in applied papers.
- In R: `vcovHC(lm_model, type="HC2")` computes this variance using $\hat{e}_i^2$ to estimate the diagonal of $\mathbf{D}$.
- Under homoskedasticity ($\mathbf{D} = \sigma^2\mathbf{I}$), the sandwich simplifies to $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ —the classical formula. Robust SEs and classical SEs agree.
- When they diverge, it signals heteroskedasticity in your data.

Special case, $\sigma_i^2 = \sigma_j^2 = \sigma^2$

- If $\mathbf{D} = \mathbf{I}_n \sigma^2$, (assumption 4*), then we call the error homoskedastic.

$$\begin{aligned} \text{var}[\hat{\beta}|\mathbf{X}] &= (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \mathbf{D} \mathbf{X} (\mathbf{X}'\mathbf{X})^{-1} \\ &= (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \mathbf{I}_n \sigma^2 \mathbf{X} (\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \mathbf{X} (\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} \end{aligned}$$

- Homoskedasticity is a convenient assumption, but it also justifies relying on OLS: under $\mathbf{D} = \sigma^2 \mathbf{I}$, Gauss-Markov guarantees no linear estimator can do better.

Proof of Gauss Markov Theorem

Formal statement: In the homoskedastic linear model, if $\tilde{\beta}$ is a linear unbiased estimator of β , then

$$\text{var}(\tilde{\beta}|\mathbf{X}) \geq \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$$

Proof (4 steps):

- 1 Let $\mathbf{A}$ be any $n \times k$ linear function of $\mathbf{X}$ such that $\mathbf{A}'\mathbf{X} = \mathbf{I}_k$.
By (A2) and (A3), $\mathbf{A}'\mathbf{Y} = \tilde{\beta}$ is an *unbiased* estimator of β , as
 $E[\tilde{\beta}|\mathbf{X}] = E[\mathbf{A}'\mathbf{Y}|\mathbf{X}] = \mathbf{A}'E[\mathbf{Y}|\mathbf{X}] = \mathbf{A}'\mathbf{X}\beta = \beta$.
- 2 Under (A1) and (A4), the variance of $\hat{\beta}$ is $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ and the variance of $\tilde{\beta}$ is $\sigma^2\mathbf{A}'\mathbf{A}$.
- 3 Evaluate $\mathbf{A}'\mathbf{A} - (\mathbf{X}'\mathbf{X})^{-1}$ to show that it is positive semi-definite.

Proof of Gauss Markov Theorem

Set $\mathbf{C} = \mathbf{A} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$, and note that $\mathbf{X}'\mathbf{C} = 0$,

$$\begin{aligned}\mathbf{A}'\mathbf{A} - (\mathbf{X}'\mathbf{X})^{-1} &= (\mathbf{C} + \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1})'(\mathbf{C} + \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}) - (\mathbf{X}'\mathbf{X})^{-1} \\ &= \mathbf{C}'\mathbf{C} + \underbrace{\mathbf{C}'\mathbf{X}}_{=0}(\mathbf{X}'\mathbf{X})^{-1} + (\mathbf{X}'\mathbf{X})^{-1}\underbrace{\mathbf{X}'\mathbf{C}}_{=0} \\ &\quad + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} - (\mathbf{X}'\mathbf{X})^{-1} \\ &= \mathbf{C}'\mathbf{C}\end{aligned}$$

4 Any matrix $\mathbf{C}'\mathbf{C}$ is positive semi-definite: $\mathbf{x}'\mathbf{C}'\mathbf{C}\mathbf{x} = \|\mathbf{C}\mathbf{x}\|^2 \geq 0$.

So $\mathbf{A}'\mathbf{A} - (\mathbf{X}'\mathbf{X})^{-1} \geq 0$ and

$$\text{var}(\tilde{\beta}|\mathbf{X}) \geq \sigma^2(\mathbf{X}'\mathbf{X})^{-1} \quad \square$$

No linear estimator will get a lower variance! OLS is the **best** among **linear unbiased** estimators (BLUE).

What Does BLUE Mean for Applied Work?

- Under homoskedasticity, OLS gives you the *tightest possible* confidence intervals among linear unbiased estimators.
- Practical implication: if $\text{var}[\mathbf{e}|\mathbf{X}] = \sigma^2 \mathbf{I}$, there is no reason to search for a cleverer estimator—OLS is already optimal.
- What BLUE does *not* guarantee:
 - If errors are heteroskedastic, OLS is no longer efficient $\rightarrow$ GLS can do better.
 - If $\mathbb{E}[\mathbf{e}|\mathbf{X}] \neq 0$ (omitted variables, simultaneity), OLS is biased regardless of efficiency.
 - BLUE says nothing about nonlinear estimators (MLE may dominate if you know the error distribution).
- Bottom line: Gauss-Markov tells you *when you can stop looking* for a better estimator, and when you cannot.

Gauss-Markov in Action: OLS vs. Alternatives

```
set.seed(42); B <- 5000; n <- 50
b_ols <- b_split <- numeric(B)
for (i in 1:B) {
  x <- rnorm(n); e <- rnorm(n, 0, 2)
  y <- 2 + 3*x + e
  b_ols[i] <- coef(lm(y ~ x))[2]
  h1 <- 1:(n/2); h2 <- (n/2+1):n
  b_split[i] <- (coef(lm(y[h1] ~ x[h1]))[2]
    + coef(lm(y[h2] ~ x[h2]))[2])/2
}
c(var(b_ols), var(b_split))
# ~ 0.08 ~ 0.16
```

- DGP: $Y = 2 + 3X + e$, $e \sim N(0, 4)$.
- Split-sample: run OLS on each half, average the slopes. Linear, unbiased—but wastes information.
- OLS variance $\approx$ half the split-sample variance.

Under homoskedasticity, OLS uses all the information in the data. Any other linear unbiased estimator wastes some.

Hansen's Gauss Markov Theorem (Technical)

In the homoskedastic linear model, if $\tilde{\beta}$ is ~~a linear~~ *any* unbiased estimator of β , then

$$\text{var}(\tilde{\beta}|\mathbf{X}) \geq \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$$

This result is derived using the Cramér-Rao bound.

Theoretical Criteria for Estimators

- Gauss Markov: OLS is "best" by reference to any alternative unbiased linear estimator.
- Cramér-Rao bound: the precision (inverse of the variance) of any unbiased estimator is bounded by the Fisher information of the estimator.
- We will define "information" more formally when we get to Maximum Likelihood Estimation.

Method of Moments Estimation

- A **method of moments estimator** (MME) sets sample moments equal to population moments and solves for the parameter.
- Population moment condition: $\mathbb{E}[g(\mathbf{x}_i, Y_i, \theta)] = 0$
- Sample analog: $\frac{1}{n} \sum_{i=1}^n g(\mathbf{x}_i, Y_i, \hat{\theta}) = 0$
- OLS is a method of moments estimator:
 - Population: $\mathbb{E}[\mathbf{x}_i(Y_i - \mathbf{x}_i'\beta)] = 0$ (from $\mathbb{E}[\mathbf{x}_i e_i] = 0$)
 - Sample: $\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i(Y_i - \mathbf{x}_i'\hat{\beta}) = 0 \Rightarrow \mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\beta}) = 0$
 - Solving gives $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ —the OLS estimator.

Method of Moments for σ^2

- Under homoskedasticity, the population moment is $\mathbb{E}[e_i^2] = \sigma^2$.
- The MME replaces e_i with $\hat{e}_i$ and averages:

$$\hat{\sigma}_{MM}^2 = \frac{1}{n} \sum_{i=1}^n \hat{e}_i^2$$

- This is biased: $\mathbb{E}[\hat{\sigma}_{MM}^2 | \mathbf{X}] = \frac{n-k}{n} \sigma^2$ (shown later in the residuals section).
- The bias-corrected version $s^2 = \frac{1}{n-k} \sum \hat{e}_i^2$ is the standard estimator in practice.
- Method of moments generalizes to **GMM** (Generalized Method of Moments) when we have more moment conditions than parameters.

Where Do Moment Conditions Come From?

Moment conditions are not statistical conveniences—they encode **substantive claims about the world**. Three major sources:

- 1 **Behavioral / equilibrium conditions.** An optimization or equilibrium model implies a first-order condition:

$$\mathbb{E}[Z_i \cdot (\text{marginal benefit} - \text{marginal cost})] = 0$$

Variables in the agent's information set must be orthogonal to the residual.

- 2 **Institutional or exogenous variation.** A design feature or natural experiment makes some variable plausibly independent of the error:

$$\mathbb{E}[Z_i e_i] = 0 \quad (\text{e.g., random assignment, world prices for a small country})$$

- 3 **Accounting or definitional identities.** Budget constraints, vote shares summing to one, or flows = stocks + changes impose restrictions without behavioral assumptions.

The Theory-to-Estimator Workflow

From Mechanism to Estimator

- 1 **Mechanism:** What process links the variables?
- 2 **Exogenous feature:** What institutional rule, randomization, or equilibrium condition creates a restriction?
- 3 **Observable implication:** What pattern must hold in the data if the mechanism and assumptions are correct?
- 4 **Moment condition:** $\mathbb{E}[g(W_i, \theta)] = 0$
- 5 **Estimator:** Solve the sample analog.

OLS and GLS Through This Lens

OLS:

- **Mechanism:** Y is a linear function of X plus noise.
- **Exogenous feature:** No omitted variable correlated with X affects Y .
- **Moment condition:** $\mathbb{E}[X_i(Y_i - X_i'\beta)] = 0$.
- **Estimator:** $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$.

GLS:

- **Mechanism:** Same linear model, but we claim to know which observations are noisier.
(GLS assumes the functional form is already correct—invariance guides that prior choice.)
- **Moment condition:** $\mathbb{E}[\Sigma^{-1/2}X_i(\Sigma^{-1/2}Y_i - \Sigma^{-1/2}X_i'\beta)] = 0$
- **Estimator:** $\hat{\beta}_{GLS} = (\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{-1}\mathbf{Y}$.

The GLS Assumption

- Switching from OLS to GLS is not just an algebraic trick—it reflects a **substantive claim about the error structure**.
- **Exogenous feature:** We observe (or can estimate) the error covariance Σ —e.g., cross-country data where richer countries have more variable outcomes.
- The claim “I know Σ ” is itself a substantive assumption—if wrong, GLS can be *worse* than OLS.
- Two things can go wrong with GLS:
 - 1 Wrong Σ : you down-weight the wrong observations $\rightarrow$ efficiency loss.
 - 2 Wrong functional form: if the process is scale-invariant, you should take logs *before* worrying about Σ (recall the invariance slide).

Spherical Errors

- The Gauss Markov Theorem assumed that $E(\mathbf{e}\mathbf{e}') = \begin{bmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{bmatrix} = \sigma^2 \mathbf{I}$
- In quadratic form, $\sigma^2 \mathbf{I}$ is the formula for a sphere

$$\begin{aligned} \mathbf{e}'(\sigma^2 \mathbf{I})\mathbf{e} &= \sigma^2 e_1^2 + \sigma^2 e_2^2 + \cdots + \sigma^2 e_n^2 = q \\ &= e_1^2 + e_2^2 + \cdots + e_n^2 = q/\sigma^2 \end{aligned}$$

Non-Spherical Errors

- More generally we think that errors may exhibit variation: If

$$E(\mathbf{ee}') = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & \sigma_N^2 \end{bmatrix}, \sigma_i \neq \sigma_j, \text{ then we say we have heteroskedasticity.}$$

- Heteroskedasticity can occur when cases are aggregates of uneven size.

$$\text{■ If } E(\mathbf{ee}') = \begin{bmatrix} \sigma^2 & a & \cdots & b \\ a & \sigma^2 & \cdots & d \\ \vdots & \vdots & \cdots & \vdots \\ b & d & \cdots & \sigma^2 \end{bmatrix}, a \neq b \neq d \neq 0, \text{ then we say we have autocorrelation.}$$

- More generally, we will allow for both.

Where Do Non-Spherical Errors Arise?

■ Heteroskedasticity:

- Cross-country regressions: richer countries often have more variable outcomes (GDP growth, trade flows).
- Earnings regressions: variance of wages increases with education and experience.
- Any regression where the outcome is bounded below (e.g. expenditure ≥ 0): variance shrinks near the bound.

■ Autocorrelation:

- Time series: an economic shock in one quarter persists into the next.
 - Panel data: unobserved country or individual traits make errors within a unit correlated.
 - Spatial data: neighboring counties share unobserved shocks (weather, policy spillovers).
- In all these cases, OLS $\hat{\beta}$ is still unbiased, but its variance is *not* $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ —classical standard errors are wrong.

Before GLS: Is the Specification Right?

- Heteroskedasticity sometimes signals a **misspecified functional form**, not just a variance problem.
- If GDP growth variance increases with GDP level, the underlying process may be *multiplicative*:

$$P_{t+1} = P_t(1 + g_t) \quad \Rightarrow \quad \Delta \log P_t \approx g_t$$

A log specification makes effects proportional—and may eliminate the heteroskedasticity.

- The key question: **what transformation leaves the effect unchanged?**
 - *Translation-invariant* mechanism (additive accumulation) $\rightarrow$ levels, $\mathbb{E}[Y | X] = X'\beta$
 - *Scale-invariant* mechanism (proportional growth) $\rightarrow$ logs, $\mathbb{E}[\log Y | X] = X'\beta$
- In a diff-in-diff design, this distinction matters directly: parallel trends in *levels* vs. parallel trends in *growth rates* are different identifying assumptions.
- GLS addresses heterogeneity that remains *after* the right specification is chosen.

Generalized Least Squares

- Consider the following linear regression model:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$$

- Allow for heteroskedasticity or correlation in the errors:

$$\mathbb{E}[\mathbf{e}|\mathbf{X}] = \mathbf{0}$$

$$\text{var}[\mathbf{e}|\mathbf{X}] = \Sigma\sigma^2$$

- Here Σ is $n \times n$ and may be a function of $\mathbf{X}$, σ^2 is the common component to the diagonals (could be 1).
- Recall, Gauss-Markov proves we can do no better than OLS when $\text{var}[\mathbf{e}|\mathbf{X}] = \sigma^2\mathbf{I}$. But now we can do better.

Derivation of Variance (again)

$$\begin{aligned} \text{var}[\hat{\beta}] &= E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'] \\ &= E[((\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}) - \beta)((\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}) - \beta)'] \\ &= E[((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e})((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e})'] \\ &= E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}\mathbf{e}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}] \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\mathbf{e}\mathbf{e}']\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\Sigma\sigma^2\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\Sigma\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \end{aligned}$$

Lower Bound on Variance of Estimators

Theorem: Given assumptions of linear model, if $\tilde{\beta}$ is an unbiased estimator of β , then

$$\text{var}[\tilde{\beta}] \geq \sigma^2(\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}$$

All we need is to know Σ^{-1} , then correct for it.

Proof: Efficiency Lower Bound for Linear Estimators (Hansen Ex. 4.6)

Theorem: If $\tilde{\beta} = \mathbf{A}'\mathbf{Y}$ is linear and unbiased, then $\text{var}(\tilde{\beta}|\mathbf{X}) \geq \sigma^2(\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}$.

Proof: Transform to the spherical model via $\Sigma^{-1/2}$:

- 1 Define $\tilde{\mathbf{Y}} = \Sigma^{-1/2}\mathbf{Y}$, $\tilde{\mathbf{X}} = \Sigma^{-1/2}\mathbf{X}$, $\tilde{\mathbf{e}} = \Sigma^{-1/2}\mathbf{e}$, so $\text{var}[\tilde{\mathbf{e}}|\mathbf{X}] = \sigma^2\mathbf{I}_n$.
- 2 Write $\tilde{\beta} = \mathbf{A}'\mathbf{Y} = \mathbf{A}'\Sigma^{1/2}\tilde{\mathbf{Y}} \equiv \tilde{\mathbf{A}}'\tilde{\mathbf{Y}}$.
- 3 Unbiasedness: $\tilde{\mathbf{A}}'\tilde{\mathbf{X}} = \mathbf{A}'\Sigma^{1/2}\Sigma^{-1/2}\mathbf{X} = \mathbf{A}'\mathbf{X} = \mathbf{I}_k$.
- 4 Apply Gauss-Markov to the transformed (spherical) model:

$$\text{var}[\tilde{\beta}|\mathbf{X}] = \sigma^2\tilde{\mathbf{A}}'\tilde{\mathbf{A}} \geq \sigma^2(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1} = \sigma^2(\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1} \quad \square$$

Intuition: $\Sigma^{-1/2}$ converts any non-spherical model into a spherical one. In the spherical world, OLS (= GLS on original data) is already BLUE.

Aside: Matrix Decomposition

- Any matrix Σ that is positive definite and symmetric can be factored:

$$\Sigma = \mathbf{C}\mathbf{\Lambda}\mathbf{C}'$$

- The columns of $\mathbf{C}$ are the eigenvectors of Σ .
- $\mathbf{\Lambda}$ is a diagonal matrix of the eigenvalues of Σ .
- $\mathbf{\Lambda}^{1/2}$ is a diagonal matrix of the square roots of the eigenvalues of Σ .

$$\Sigma = \mathbf{C}\mathbf{\Lambda}\mathbf{C}' = \mathbf{C}\mathbf{\Lambda}^{1/2}\mathbf{\Lambda}^{1/2}\mathbf{C}' = \mathbf{T}\mathbf{T}'$$

$$\Sigma^{-1} = \mathbf{C}\mathbf{\Lambda}^{-1}\mathbf{C}' = \mathbf{C}\mathbf{\Lambda}^{-1/2}\mathbf{\Lambda}^{-1/2}\mathbf{C}' = \Sigma^{-1/2}(\Sigma^{-1/2})'$$

Aitken (1935) Generalized Least Squares

- If we know Σ , we can do no better than to premultiply our linear model with $\Sigma^{-1/2}$,

$$\begin{aligned}\mathbf{y} &= \mathbf{X}\boldsymbol{\beta} + \mathbf{e} \\ \Sigma^{-1/2}\mathbf{y} &= \Sigma^{-1/2}\mathbf{X}\boldsymbol{\beta} + \Sigma^{-1/2}\mathbf{e} \\ \tilde{\mathbf{y}} &= \tilde{\mathbf{X}}\boldsymbol{\beta} + \tilde{\mathbf{e}} \\ \tilde{\boldsymbol{\beta}}_{GLS} &= (\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}'\tilde{\mathbf{y}} \\ &= ((\Sigma^{-1/2}\mathbf{X})'(\Sigma^{-1/2}\mathbf{X}))^{-1}(\Sigma^{-1/2}\mathbf{X})'(\Sigma^{-1/2}\mathbf{y}) \\ &= (\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{-1}\mathbf{y}\end{aligned}$$

Here we have $E[\tilde{\boldsymbol{\beta}}_{GLS}] = \boldsymbol{\beta}$ and $\text{var}(\tilde{\boldsymbol{\beta}}_{GLS}) = \sigma^2(\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}$

GLS is BLUE

- Suppose b is an alternative linear unbiased estimator that differs by A

$$\begin{aligned} b &= [(X' \Sigma^{-1} X)^{-1} X' \Sigma^{-1} + A] Y \\ &= [(X' \Sigma^{-1} X)^{-1} X' \Sigma^{-1} + A] (X\beta + e) \\ &= [(X' \Sigma^{-1} X)^{-1} X' \Sigma^{-1} + A] X\beta + [(X' \Sigma^{-1} X)^{-1} X' \Sigma^{-1} + A] e \end{aligned}$$

$$AX = 0 \quad (\text{Because } b \text{ is unbiased.})$$

$$\begin{aligned} \text{var}(b) &= \sigma^2 [(X' \Sigma^{-1} X)^{-1} X' \Sigma^{-1} + A] \Sigma [(X' \Sigma^{-1} X)^{-1} X' \Sigma^{-1} + A]' \\ &= \sigma^2 [(X' \Sigma^{-1} X)^{-1} + A \Sigma A' + (X' \Sigma^{-1} X)^{-1} X' A' + AX (X' \Sigma^{-1} X)^{-1}] \\ &= \sigma^2 [(X' \Sigma^{-1} X)^{-1} + A \Sigma A' + 0 + 0] \\ &= \sigma^2 (X' \Sigma^{-1} X)^{-1} + \sigma^2 A \Sigma A' \end{aligned}$$

- Σ is positive definite, so $\sigma^2 A \Sigma A' \geq 0$; minimum variance requires $A = 0$



GLS Properties

- GLS is unbiased ($\mathbb{E}[\hat{\beta}_{GLS}|\mathbf{X}] = \beta$) and BLUE under the correct Σ .
- Algebraically, GLS is OLS on the transformed data $(\tilde{\mathbf{y}}, \tilde{\mathbf{X}})$. The only structural difference is that the GLS “hat matrix” $\mathbf{X}(\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{-1}$ is not symmetric.
- **Feasibility problem:** GLS requires knowing Σ , which is rarely available in practice.
- **Solution:** Feasible GLS (FGLS) estimates Σ in a first step (e.g. from OLS residuals), then applies GLS with $\hat{\Sigma}$. Asymptotically equivalent to GLS under regularity conditions.

When GLS Beats OLS: A Two-Group Example

- Survey data from two groups: Group A (rich, $n_A = 50$, $\sigma_A^2 = 100$) and Group B (poor, $n_B = 50$, $\sigma_B^2 = 1$).
- OLS gives *equal weight* to every observation $\rightarrow$ noisy Group A observations inflate variance.
- GLS (= WLS here) weights by $w_i = 1/\sigma_i^2$: Group A gets weight 1/100, Group B gets weight 1.

Variance comparison (scalar case, equal group sizes):

$$\text{var}(\hat{\beta}_{OLS}) \propto \frac{\sigma_A^2 + \sigma_B^2}{2} = \frac{101}{2} \quad \text{var}(\hat{\beta}_{GLS}) \propto \frac{1}{1/\sigma_A^2 + 1/\sigma_B^2} = \frac{100}{101} \approx 1$$

GLS is just common sense: trust precise observations more. When you know which observations are noisier, use that information.

Comparing OLS and WLS: R Example

```
set.seed(99); B <- 5000; n <- 100
b_ols <- b_wls <- numeric(B)

for (i in 1:B) {
  x <- rnorm(n)
  # Two groups: first 50 noisy, last 50 precise
  sigma <- c(rep(10, 50), rep(1, 50))
  y <- 1 + 2*x + rnorm(n, 0, sigma)

  b_ols[i] <- coef(lm(y ~ x))[2]
  b_wls[i] <- coef(lm(y ~ x, weights = 1/sigma^2))[2]
}

c(sd(b_ols), sd(b_wls)) # WLS SE is smaller
# ~ 1.01 ~ 0.14
```

- Both estimators are unbiased ($\hat{\beta} \approx 2$), but WLS standard errors are $\sim 7\times$ smaller.
- In practice, use `lm(..., weights = 1/sigma_hat^2)` when group variances are known or estimable.

Residuals

- We will be using estimates of the residuals to construct covariance matrix estimators that do not require homoskedasticity.
- The residuals $\hat{e}_i = Y_i - \mathbf{x}_i'\hat{\beta}$ can be written in vector notation as $\hat{\mathbf{e}} = \mathbf{M}\mathbf{e}$.
- Recall $\mathbf{M} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$
- Given $\mathbf{X}$, the expectation of the residuals is zero :

$$\mathbb{E}[\hat{\mathbf{e}}|\mathbf{X}] = \mathbb{E}[\mathbf{M}\mathbf{e}|\mathbf{X}] = \mathbf{M}\mathbb{E}[\mathbf{e}|\mathbf{X}] = \mathbf{0}$$

- Given $\mathbf{X}$, the variance of the residuals is:

$$\text{var}[\hat{\mathbf{e}}|\mathbf{X}] = \text{var}[\mathbf{M}\mathbf{e}|\mathbf{X}] = \mathbf{M}\text{var}[\mathbf{e}|\mathbf{X}]\mathbf{M} = \mathbf{MDM}$$

Homoskedastic Errors

- If we assume homoskedasticity, $\mathbf{E}[e^2|X] = \sigma^2$

$$\text{var}[\hat{\mathbf{e}}|\mathbf{X}] = \text{var}[\mathbf{M}\mathbf{e}|\mathbf{X}] = \mathbf{M}\text{var}[\mathbf{e}|\mathbf{X}]\mathbf{M} = \mathbf{M}\mathbf{I}\sigma^2\mathbf{M} = \mathbf{M}\sigma^2$$

- Note that the i th diagonal element of $\mathbf{M}$ is $1 - h_{ii}$, so

$$\text{var}[\hat{e}_i|\mathbf{X}] = \mathbb{E}[\hat{e}_i^2|\mathbb{X}] = (1 - h_{ii})\sigma^2 \neq \sigma^2$$

- $\hat{e}_i^2$ is a biased estimator and it is heteroskedastic.

Prediction Errors

- The prediction errors $\tilde{e}_i = Y_i - \mathbf{x}_i \hat{\beta}_{-i} = (1 - h_{ii})^{-1} \hat{e}_i$.
- We defined $\mathbf{M}^* = \text{diag}\{(1 - h_{11})^{-1}, (1 - h_{22})^{-1}, \dots, (1 - h_{nn})^{-1}\}$.
- $\tilde{\mathbf{e}} = \mathbf{M}^* \hat{\mathbf{e}} = \mathbf{M}^* \mathbf{M} \mathbf{e}$
- Given $\mathbf{X}$, the expectation of the prediction errors is zero :

$$\mathbb{E}[\tilde{\mathbf{e}}|\mathbf{X}] = \mathbf{M}^* \mathbf{M} \mathbb{E}[\mathbf{e}|\mathbf{X}] = \mathbf{0}$$

- Given $\mathbf{X}$, the variance of the prediction errors is:

$$\text{var}[\tilde{\mathbf{e}}|\mathbf{X}] = \text{var}[\mathbf{M}^* \mathbf{M} \mathbf{e}|\mathbf{X}] = \mathbf{M}^* \mathbf{M} \text{var}[\mathbf{e}|\mathbf{X}] \mathbf{M} \mathbf{M}^* = \mathbf{M}^* \mathbf{M} \mathbf{D} \mathbf{M} \mathbf{M}^*$$

Prediction Errors under homoskedasticity

- Under homoskedasticity, the variance of the prediction errors is:

$$\text{var}[\tilde{\mathbf{e}}|\mathbf{X}] = \text{var}[\mathbf{M}^* \mathbf{M} \mathbf{e}|\mathbf{X}] = \mathbf{M}^* \mathbf{M} \text{var}[\mathbf{e}|\mathbf{X}] \mathbf{M} \mathbf{M}^* = \mathbf{M}^* \mathbf{M} \sigma^2 \mathbf{M} \mathbf{M}^* = \mathbf{M}^* \mathbf{M} \mathbf{M}^* \sigma^2$$

- The variance of the i th prediction error is then:

$$\begin{aligned} \text{var}[\tilde{e}_i|\mathbf{X}] &= \mathbb{E}[\tilde{e}_i^2|\mathbf{X}] \\ &= (1 - h_{ii})^{-1} (1 - h_{ii}) (1 - h_{ii})^{-1} \sigma^2 \\ &= (1 - h_{ii})^{-1} \sigma^2 \end{aligned}$$

- $\sum \tilde{e}_i^2 = \sum ((1 - h_{ii})^{-1} \hat{e}_i)^2 = \text{PRESS}$, "predictive error".

Standardized Residuals

- $\text{var}[\hat{e}_i|\mathbf{X}] = (1 - h_{ii})\sigma^2$ varies with $\mathbf{X}$
- To make it constant, we can scale the residuals by $(1 - h_{ii})^{-1/2}$

$$\bar{e}_i = (1 - h_{ii})^{-1/2} \hat{e}_i$$

- $\bar{\mathbf{e}} = \mathbf{M}^{*1/2} \mathbf{M} \mathbf{e}$
- $\text{var}[\bar{e}_i|\mathbf{X}] = \mathbb{E}[\bar{e}_i^2|\mathbf{X}] = \sigma^2$
- If the error is homoskedastic, $\bar{e}_i$ has the same bias and variance as the original errors.
- If the error is heteroskedastic, these standardized residuals are not.
- In R, these are recovered by “`rstandard(lmmod)`”: sometimes compared to -2 , 2 .

Estimating $\hat{\sigma}^2$, s^2 , $\bar{\sigma}^2$

- The error variance σ^2 measures the unexplained part of the regression.
- Three estimators:
 - 1 The method of moments estimator is $\hat{\sigma}^2 = \frac{1}{n} \sum \hat{e}_i^2$ (Biased)
 - 2 The bias-corrected estimator is $s^2 = \frac{1}{n-k} \sum \hat{e}_i^2$ (sigma in R.)
 - 3 The standardized estimator $\bar{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \bar{e}_i^2 = \frac{1}{n} \sum_{i=1}^n (1 - h_{ii})^{-1} \hat{e}_i^2$

Estimating $\hat{\sigma}^2$

- Estimator (1) $\hat{\sigma}^2$ takes the average of the squared residuals: $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \hat{e}^2$
- The expectation of the estimator uses the following fact:

$$\hat{\sigma}^2 = \frac{1}{n} \mathbf{e}' \mathbf{M} \mathbf{e} = \frac{1}{n} \text{tr}(\mathbf{e}' \mathbf{M} \mathbf{e}) = \frac{1}{n} \text{tr}(\mathbf{M} \mathbf{e} \mathbf{e}')$$

$$\begin{aligned} \mathbb{E}[\hat{\sigma}^2 | \mathbf{X}] &= \frac{1}{n} \text{tr}(\mathbb{E}[\mathbf{M} \mathbf{e} \mathbf{e}' | \mathbf{X}]) \\ &= \frac{1}{n} \text{tr}(\mathbf{M} \mathbb{E}[\mathbf{e} \mathbf{e}' | \mathbf{X}]) \\ &= \frac{1}{n} \text{tr}(\mathbf{M} \mathbf{D}) \\ &= \frac{1}{n} \sum_{i=1}^n (1 - h_{ii}) \sigma_i^2 \end{aligned}$$

Estimating $\hat{\sigma}^2$

- Under conditional homoskedasticity

$$\begin{aligned}\mathbb{E}[\hat{\sigma}^2|\mathbf{X}] &= \frac{1}{n} \text{tr}(\mathbb{E}[\mathbf{M}\mathbf{e}\mathbf{e}'|\mathbf{X}]) \\ &= \frac{1}{n} \text{tr}(\mathbf{M}\mathbb{E}[\mathbf{e}\mathbf{e}'|\mathbf{X}]) \\ &= \frac{1}{n} \text{tr}(\mathbf{M}\mathbf{I}_n\sigma^2) \\ &= \sigma^2 \frac{n-k}{n}\end{aligned}$$

What should I report?

- The bias-corrected error variance estimator s^2 is used throughout applied work.
- It is used to calculate standard errors, F-tests, t-test, and confidence intervals.
- It is typically reported as the $\text{RMSE} = \sqrt{s^2}$
- However, s^2 assumes homoskedasticity:
 - You will use robust sandwich estimators like HC2 and HC3 (more on that later),
 - you can report $\bar{\sigma}^2$, which is unbiased under heteroskedasticity.